

ANDRÉ ROGÉRIO DA SILVA FILHO

Software Consultant & AI Engineer | Full-Stack · Autonomous Agents · Applied AI & Research

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SUMMARY

Software Consultant, AI Engineer, and Systems Researcher specializing in full-stack architecture, autonomous AI agents, and cyber-physical systems. Proven track record designing and delivering end-to-end production platforms using TypeScript, Python, React, Next.js, Node.js, and PostgreSQL. At CBI, acts as technical consultant leading enterprise software engineering, autonomous AI agent architecture (Google Gemini, MCP servers, advanced RAG with hybrid search), vendor partnerships for software development, automation, and systems maintenance (including Google), GCP cloud governance, and identity compliance (89% overall and 91% IAM score in Intune). In research and residency (UniFacens/MCTI/Softex), engineers industrial Digital Twins over Private 5G networks and cognitive memory architectures with 3D spatial interfaces (Memory Lens), targeting international publication.

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, Node.js, React 18/19, Next.js 16, Express, Fastify, PyTest, Three.js, React Three Fiber

Applied AI & MLOps: Autonomous AI Agents (LangGraph, MCP Servers), LLMs (Google Gemini API), Advanced RAG (Hybrid Search & Reranking), Document Intelligence & Data Pipelines, Vector Embeddings, Confined AI, Structured Outputs (JSON Schema), AI Evaluation & Testing, Zod

Cloud, DevOps & Data: Google Cloud Platform, Microsoft Azure (Entra ID, Graph API), Docker, Kubernetes (K8s), CI/CD Pipelines, MLOps, PostgreSQL, Redis, Supabase, Prisma ORM, Git

Enterprise & Security: Responsible AI & Governance, Certified AppSec Practitioner (CAP), AES-256-GCM Encryption, LGPD/GDPR, Private 5G Networks, Intune, M365

PROJECTS

Memory Lens | Cognitive Computing & Spatial Interface

[GitHub](#)

TypeScript, React 19, Three.js, React Three Fiber, WebGL/GLSL, Express, Firebase, Gemini API, Vitest

- Designed and built a cognitive memory platform modeling human memory retention, associative recall, and structured forgetting using vector embeddings, episodic clustering, and Ebbinghaus-based exponential decay, reducing token storage by up to 80%.
- Engineered a 3D spatial interface ("Chief Organism") in React Three Fiber with custom GLSL shaders synchronized to system state via an event bus.
- Built 16 REST API endpoints with Gemini structured JSON schemas and IDOR security test coverage in Vitest. Evolved into formal academic research under Prof. Dr. Fabricio Londero at UniFacens and is now being developed into an academic paper.

FundiVR | Industrial Digital Twin & VR Training Simulator

[GitHub](#)

React 18, TypeScript, Fastify, PostgreSQL, Prisma ORM, WebSockets, Docker, Vitest

- Architected a full-stack digital twin simulator for training aluminum furnace operators in 700°C+ environments, with operational state machines covering PPE checks, thermal control, skimming, and emergency shutdown procedures.
- Engineered real-time bi-directional telemetry over WebSockets in a modular pnpm monorepo, with an automated scoring engine evaluating precision, reaction time, and SOP compliance.
- Built a decoupled zero-dependency demo mode enabling standalone operation without database connectivity.

Enterprise ITSM & AI Assistant

[GitHub](#)

Next.js 16, TypeScript, Supabase, PostgreSQL, Gemini 2.5 Flash, Microsoft Graph API, Redis, Zod

- Built an ITSM platform with intelligent ticket deflection, automated triage, and SLA tracking, piloted with 22 internal users.
- Implemented a Confined AI pattern: LLM prompt → typed JSON output → Zod schema validation → fallback business rules → application action.
- Integrated Microsoft Graph API for real-time Intune device health and compliance telemetry during ticket intake.

Onboarding OS | Multi-Tenant SaaS

[GitHub](#)

React 19, TypeScript, Node.js, Express, PostgreSQL, Supabase, Redis, Zod

- Designed a multi-tenant platform featuring document intelligence and document-to-data processing pipelines across HR and IT, with PostgreSQL Row-Level Security for tenant isolation and AES-256-GCM document encryption.
- Built scheduled background workers for LGPD data lifecycle compliance, automatically anonymizing sensitive records upon policy expiration.

PROFESSIONAL EXPERIENCE

CBI (Companhia Brasileira de Infraestrutura) <i>Software Consultant & AI Engineer São Paulo, Brazil</i>	06/2026 - Present
- Act as technical consultant leading enterprise software architecture and engineering autonomous AI systems for mission-critical operations. - Architected and deployed "Isa", an enterprise AI agent utilizing Google Gemini, autonomous tool execution via MCP Servers, contextual RAG retrieval, and deterministic JSON Schema validation. - Architected and manage cloud infrastructure on Google Cloud Platform (GCP), orchestrating microservices on Cloud Run, Cloud SQL databases, and Secret Manager governance. - Lead technical and strategic partnerships with technology vendors for software development, automation, and systems maintenance (including Google); implement identity controls and endpoint compliance via Microsoft Intune and Graph API (89% overall and 91% IAM posture). - Provide executive consulting and lead corporate AI enablement workshops, training cross-functional business and technical teams on secure, high-impact generative AI workflows.	

Farmamed Drogarias <i>Full-Stack Developer & Automation Sorocaba, Brazil</i>	10/2025 - 03/2026
- Developed Python microservices and automation scripts for multi-site operational data consolidation, log analytics, and systems synchronization. - Built automated pipelines for enterprise identity provisioning and endpoint governance across Entra ID, Microsoft Intune, and Active Directory. - Designed fault-tolerant monitoring workflows and deterministic automation to resolve incidents and maintain high-availability SLAs.	

Wobben Windpower <i>Software & IT Systems Intern Sorocaba, Brazil</i>	11/2023 - 04/2025
- Selected for the global Microsoft Copilot Flight Crew program, evaluating generative AI enterprise features and real-world adoption workflows. - Developed internal automation scripts and integration routines for SAP ERP, Active Directory, and Microsoft 365. - Provided tier-2 technical resolution for corporate service desk incidents via ServiceNow under strict SLA benchmarks.	

EDUCATION

Higher Technology Degree in Systems Analysis and Development <i>Centro Universitário SENAC, Brazil</i>	2026
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SCIENTIFIC RESEARCH & TECHNOLOGY RESIDENCY

AI & Cyber-Physical Systems Researcher Digital Twin & 5G Residency <i>UniFacens Softex Ministry of Science, Technology and Innovation (MCTI) / MobLab 2.0 (Mover Program)</i>	2025 - 2026
- Selected through a competitive national process (360h) for advanced R&D in AI/ML, Industrial IoT, Cyber-Physical Systems, and Private 5G network architecture. - Architected and built FundiVR, an industrial digital twin training simulator featuring real-time bi-directional telemetry via WebSockets and operational safety state machines, defended before an academic and industrial jury.	

Cognitive Computing & Spatial Interfaces Researcher <i>UniFacens Advisor: Prof. Dr. Fabricio Tonetto Londero</i>	2025 - Present
- Conducting scientific research on Memory Lens, mathematically modeling memory retention, associative recall, and structured forgetting with Ebbinghaus exponential decay, vector embeddings, and knowledge graphs (reducing context token overhead by up to 80%). - Engineered an interactive 3D spatial interface in React Three Fiber with custom WebGL/GLSL shaders; authoring an academic paper targeting international peer-reviewed conference publication (ACDSA).	

CERTIFICATIONS & LANGUAGES

Certifications: Google Cloud Generative AI Leader Certified AppSec Practitioner (CAP) CCNA 1: Introduction to Networks (Cisco) Network Security & Endpoint Security (Cisco)	
Languages: English (C1+) Spanish (Intermediate / Technical Communication) Portuguese (Native)	